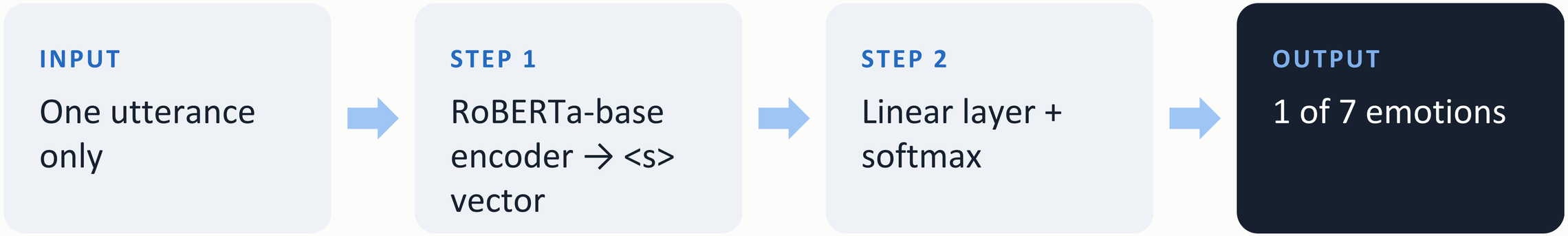
EMOTIONSHIFT · WORKSTREAM #4

RoBERTa, utterance only

How far does a pre-trained transformer get from one utterance, with no context?

KL Y · EE6405 Group Project, Table 4

How B3 works



$$p(e|u) = \mathrm{softmax}(W \cdot h_{<s>}(u) + b), \quad L = -\log p(e^*|u)$$

Whole network fine-tuned with plain cross-entropy, lr 1e-5, 125M parameters. No class weights, so disgust and fear are almost never predicted.

The core in code

```
df = load("meld")      # shared splits
m = RobertaUtterance(RobertaConfig())
# inside RobertaUtterance.fit:
args = TrainingArguments(
  num_train_epochs=cfg.epochs,
  learning_rate=cfg.lr,       # 1e-5
  eval_strategy="epoch",
  save_strategy="epoch",
  load_best_model_at_end=True,
  metric_for_best_model=cfg.select_on,
  fp16=self.device == "cuda")
m.predict(["Oh my God!", "Yeah."])
```

Best epoch chosen on dev
Scored on dev after every epoch; test used once per seed, 5 seeds.

Cost
125M parameters, 502 MB, 106 ms per utterance on CPU (demo budget 300 ms)

emotionshift/models/roberta.py
notebooks/b3_roberta.ipynb

Test results

MELD · WEIGHTED-F1

62.9

95% CI 60.0–64.8 · mean of 5 seeds

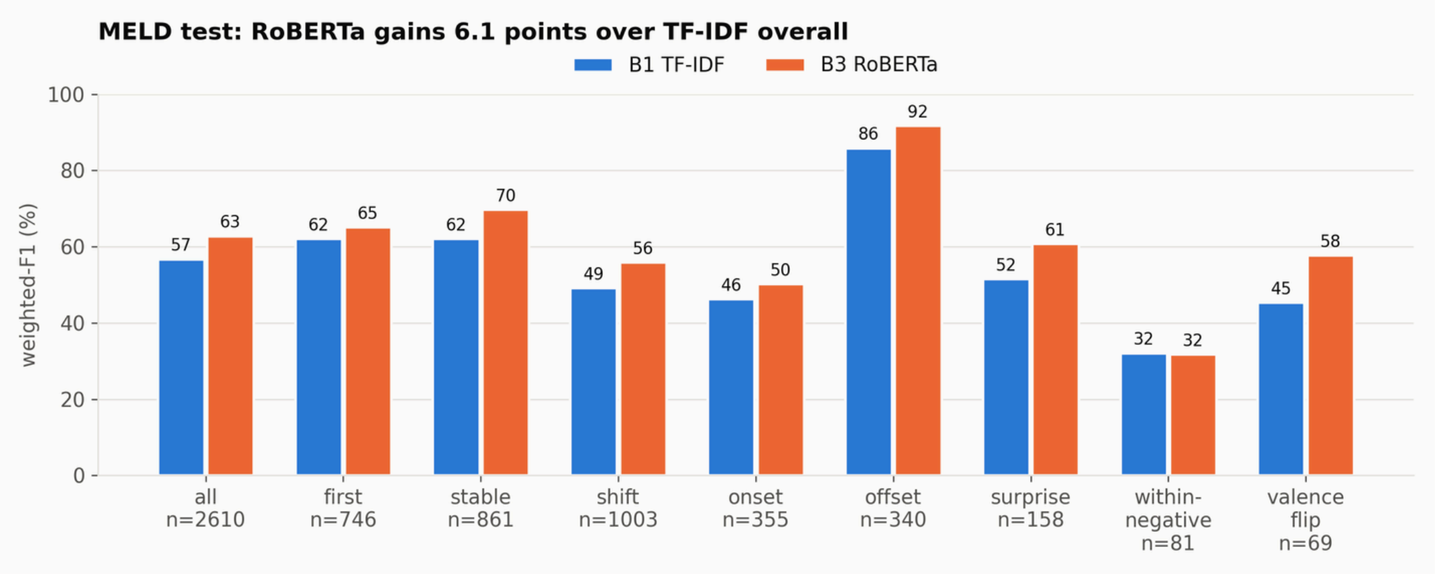
DAILYDIALOG · MICRO-F1, NO NEUTRAL

55.3

95% CI 52.5–58.2 · test_dedup, 5 seeds

MELD test, weighted-F1	**Score**
B1 TF-IDF + SVM	56.8
B3 RoBERTa (mine)	**62.9**
B2 BiGRU	55.1

Shift turns stay 14 points harder



13.8

points lower on shift turns than stable ones (55.9 vs 69.7 weighted-F1, MELD test)

B3 relies on emotion words (Integrated Gradients: wow, sorry, stupid). Within-negative shifts stay at ~32 for B1 and B3: when words don't mark the change, it must come from context.

References

APA style, alphabetical; these are merged into one group page at the end of the video.
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